$M_{\rm enc}(r_{1/2})$ Distributions from Modified Power Spectra

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1 Overview

This note describes the MassEncCDF.py pipeline that computes how modifications to the primordial matter power spectrum propagate through the halo mass function, concentration–mass relation, and NFW density profiles to produce observable differences in the enclosed mass within the stellar half-light radius, $M_{\rm enc}(r_{1/2})$. This code is adapted from a Jupyter notebook that Daniel Gilman wrote during the Spec-S5 dark matter workshop in March 2026.

Four beyond-ΛCDM scenarios are compared to a Planck 2018 ΛCDM reference:

- **WDM** — a warm dark matter–like cutoff, $P(k) \to P_{\Lambda\rm CDM}(k)\,[1+(k/k_{\rm cut})^{\nu}]^{-5/\nu}$ with $k_{\rm cut} = 150\ h/{\rm Mpc}$ and $\nu = 1.12$.
- **vEDE** — a very early dark energy (vEDE) model [1] with parameters $\mathcal{R} = 60$ and $\log_{10} a_c = -6.92$ (see Figure 9 of that paper), with all other parameters held at their Planck 2018 values.
- **AK1** — the fiducial two-field axion-kination model [2] (Figure 12 of that paper), whose $P(k)$ is provided as numerical arrays together with a ΛCDM reference so that $T(k) = \sqrt{P(k)/P_{\rm ref}(k)}$ can be applied to the Planck 2018 spectrum.
- **CDM** — the ΛCDM baseline (Planck 2018 cosmology).

All cosmological computations (power spectra, $\sigma(M)$, mass functions, and concentrations) use the Colossus toolkit with the Planck 2018 parameter set [3].

2 Power Spectra and Variance

The code begins with the Planck 2018 ΛCDM linear matter power spectrum $P_{\Lambda\rm CDM}(k)$ tabulated on a grid of 3000 points uniformly spaced in $\log_{10} k$ from 10^{-5} to $10^{3.7}\ h/{\rm Mpc}$. Each beyond-ΛCDM model modifies this spectrum through a transfer function,

$$P_X(k) = T_X^2(k)\, P_{\Lambda\rm CDM}(k)\,, \tag{1}$$

where $T_X(k)$ depends on the model. The variance of the linear density field smoothed on scale R is

$$\sigma^2(R) = \frac{1}{2\pi^2}\int_0^\infty k^2\, P(k)\, |\tilde{W}(kR)|^2\, dk\,, \tag{2}$$

with R the Lagrangian radius corresponding to halo mass M via $M = (4\pi/3)\,\bar{\rho}\,R^3$.

Figure 1 shows the four power spectra and Figure 2 shows the resulting $\sigma(M)$.

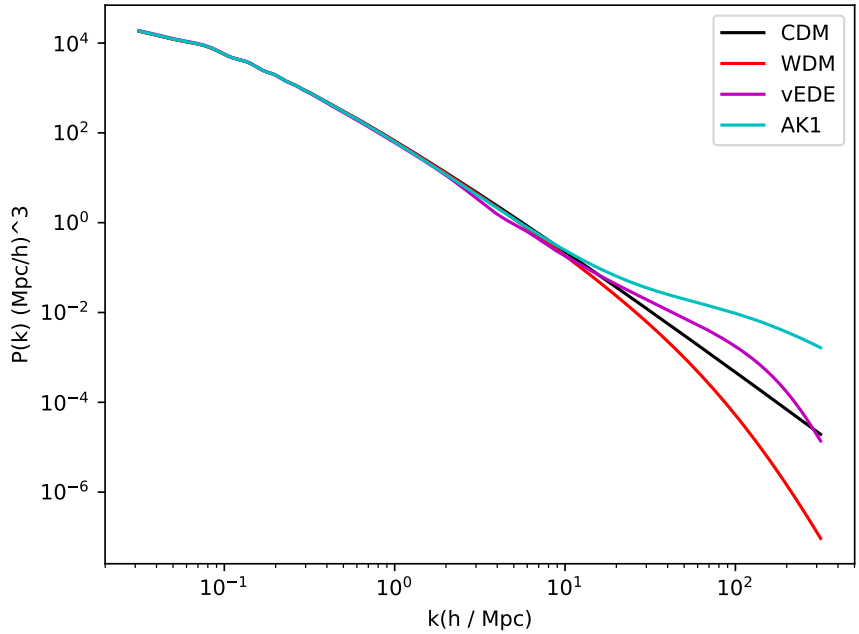

Figure 1: Linear matter power spectra for each model.

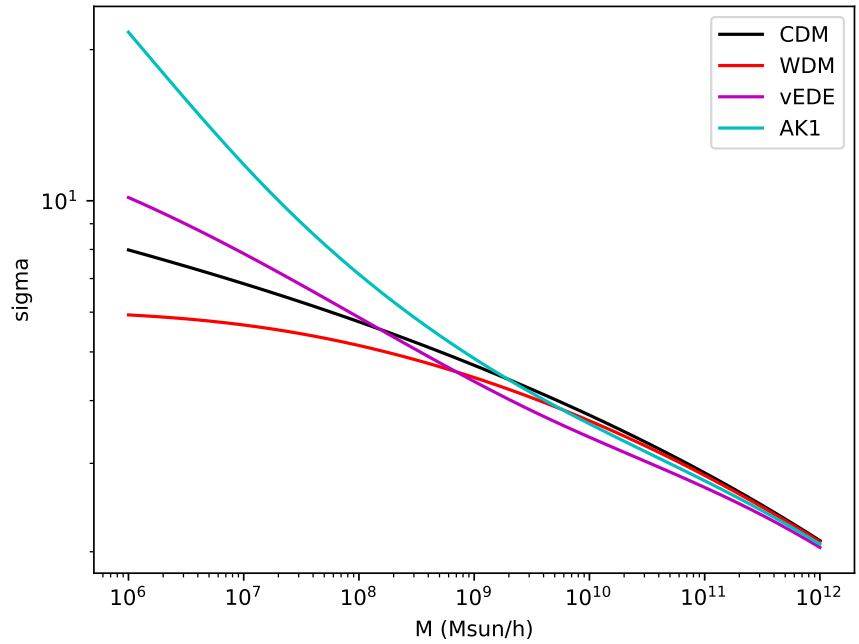

Figure 2: RMS density fluctuation $\sigma(M)$ as a function of halo mass for each model.

3 Concentration–Mass Relation

Halo concentrations are computed using the Diemer & Joyce 2019 [4] model as implemented in Colossus, evaluated at $z = 0$ for each modified power spectrum. The resulting concentration–mass relations are shown in Figure 3.

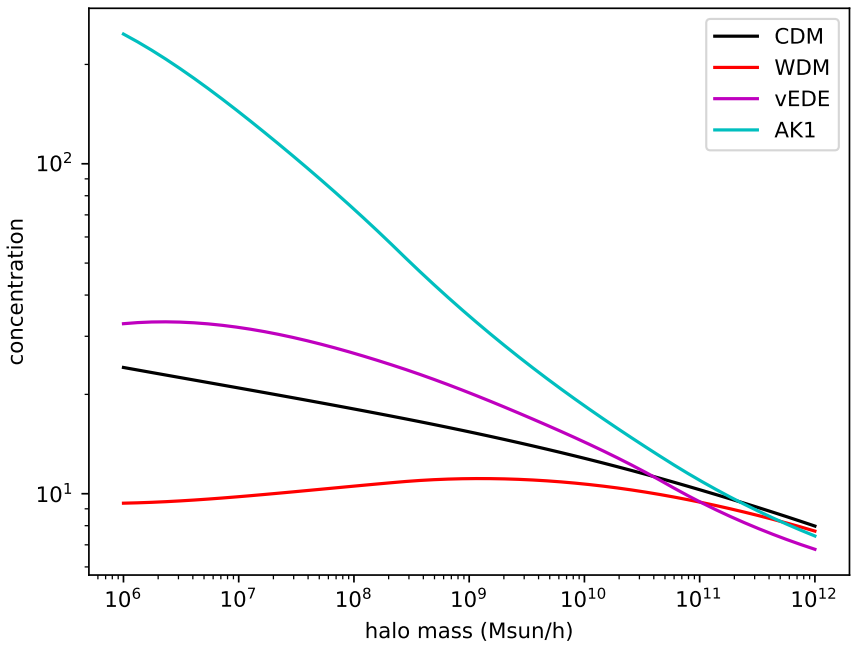

Figure 3: Concentration–mass relation $c(M)$ at $z = 0$ for each model.

4 Halo Mass Function

The halo mass function $dn/d\ln M$ is evaluated at $z = 0$ on a grid of 50 masses uniformly spaced in $\log_{10} M$ from 10^6 to 10^{12} $M_\odot/h$, using the Colossus `massFunction` routine with each model's custom power spectrum (Figure 4).

5 Monte Carlo Sampling

To generate mock halo populations we draw masses from each mass function via inverse-CDF sampling. The procedure is:

1. **Build the CDF.** Given the tabulated $f_i \equiv (dn/d\ln M)_i$ on the $\log_{10} M$ grid, define trapezoidal bin weights
$$w_i = \tfrac{1}{2}\big(f_i + f_{i+1}\big), \qquad i = 0, \ldots, N-2\,, \tag{3}$$
and set $\mathrm{CDF}_0 = 0$, $\mathrm{CDF}_j = \sum_{i=0}^{j-1} w_i$ for $j = 1, \ldots, N-1$. Because the grid spacing $\Delta\log_{10} M$ is constant it cancels after dividing by CDF_{N-1}, leaving a properly normalized CDF.

2. **Invert.** Construct a 1-D interpolant $\mathrm{CDF} \to \log_{10} M$. Draw N_{draw} uniform random variates $u \in [0, 1]$ and compute $M = 10^{\mathrm{CDF}^{-1}(u)}$.

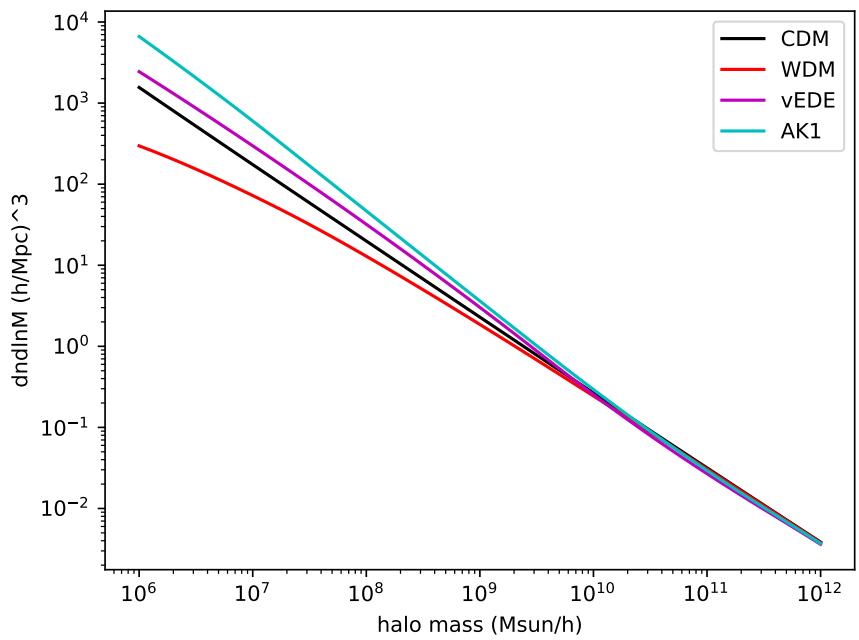

Figure 4: Halo mass function $dn/d\ln M$ at $z = 0$ for each model.

3. **Scale the number of draws.** The CDF is normalized to unity regardless of the overall amplitude of the mass function, so the number of draws must encode the relative abundance of halos across models. We set

$$N_{\mathrm{draw},X} = N_{\mathrm{draw,CDM}} \times \frac{\int (dn/d\ln M)_X \, d\log_{10} M}{\int (dn/d\ln M)_{\mathrm{CDM}} \, d\log_{10} M} \,. \tag{4}$$

The common factor of $\ln 10$ that converts $d\log_{10} M \to d\ln M$ cancels in the ratio, so integrating over $\log_{10} M$ is sufficient. Both the numerator and denominator are evaluated with `np.trapz`, matching the trapezoidal rule used for the CDF.

Figure 5 shows a histogram of the sampled masses for a diagnostic run with 5×10^7 reference draws.

6 NFW Enclosed Mass within Stellar Half-light Radius

For each sampled halo of mass M the pipeline:

1. Assigns a concentration c by interpolating the tabulated $c(M)$ relation and adding lognormal scatter with $\sigma_{\ln c} = 0.15$.

2. Computes a stellar half-light radius $r_{1/2} = 0.040\,(r_{\mathrm{vir}}/10\ \mathrm{kpc})\,\mathrm{kpc}$ [5] with lognormal scatter $\sigma_{\ln r_{1/2}} = 0.6$.

3. Evaluates the NFW enclosed mass

$$M_{\mathrm{enc}}(r_{1/2}) = 4\pi\,\rho_s\,r_s^3 \left[\ln\!\left(1 + \frac{r_{1/2}}{r_s}\right) - \frac{r_{1/2}/r_s}{1 + r_{1/2}/r_s}\right], \tag{5}$$

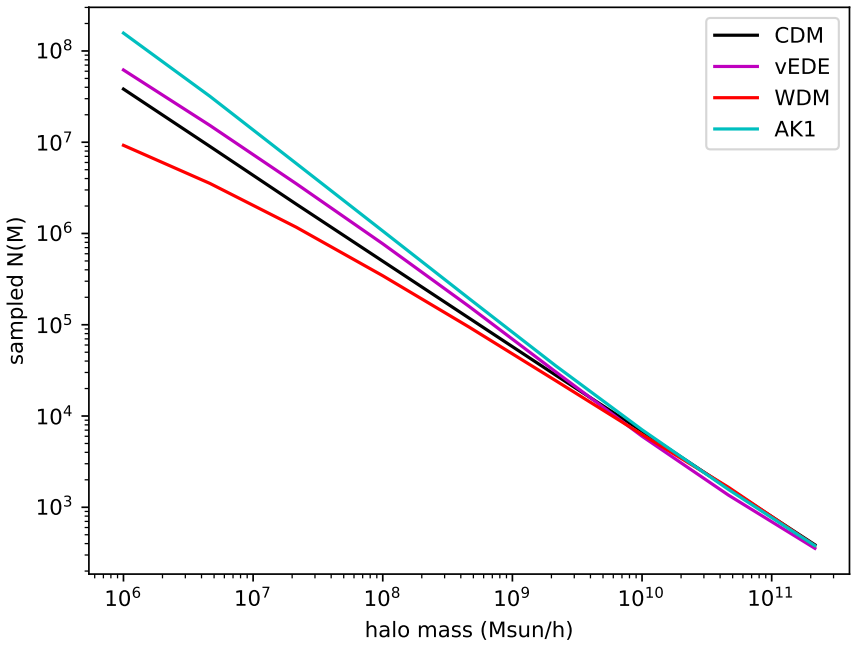

Figure 5: Histogram of halo masses drawn from each model's mass function, confirming that the sampling reproduces the correct relative abundances.

where ρ_s and r_s are the NFW scale density and scale radius obtained from (M, c) via Colossus.

Halos below $M_{\rm min} = 10^8\ M_\odot/h$ or with $r_{1/2} < 0.01\,$kpc are discarded. The final production run uses 10^8 reference CDM draws (scaled per Eq. 4 for the other models). Figure 6 shows the resulting $M_{\rm enc}/M$ versus $r_{1/2}$ scatter plot for a randomly selected subset of the sample.

7 Complementary CDF of Enclosed Mass

The main observable is the complementary cumulative distribution function (CCDF) of $M_{\rm enc}(r_{1/2})$. This is computed in 50 logarithmically spaced bins from 10^4 to $10^{10}\ M_\odot$. Because the number of draws is scaled by the mass-function integral ratio (Eq. 4), differences between models in $N(> M_{\rm enc})$ directly reflect the physical differences in halo abundance and internal structure. The result is shown in Figure 7.

References

[1] A. C. Sobotka, A. L. Erickcek and T. L. Smith, "Signatures of very early dark energy in the matter power spectrum," Phys. Rev. D **111**, no.12, 123522 (2025) doi:10.1103/9bd9-fzwh [arXiv:2409.06778 [astro-ph.CO]].

[2] R. T. Co, N. Fernandez, A. Ghalsasi, K. Harigaya and J. Shelton, JCAP **02**, 082 (2026) doi:10.1088/1475-7516/2026/02/082 [arXiv:2510.01308 [hep-ph]].

[3] B. Diemer, Astrophys. J. Suppl. **239**, no.2, 35 (2018) doi:10.3847/1538-4365/aaee8c [arXiv:1712.04512 [astro-ph.CO]].

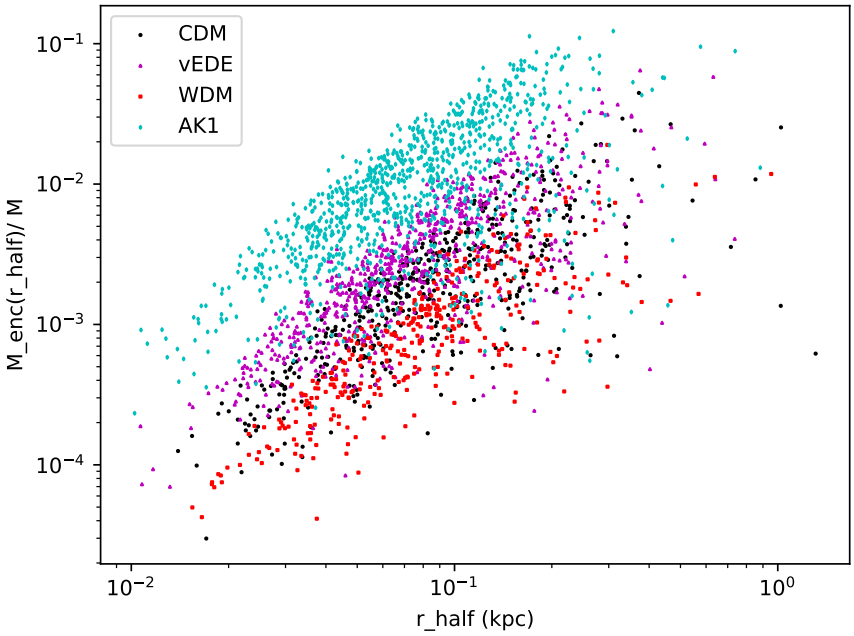

Figure 6: Enclosed mass fraction $M_{\rm enc}(r_{1/2})/M$ versus half-light radius for a random subsample of halos in each model.

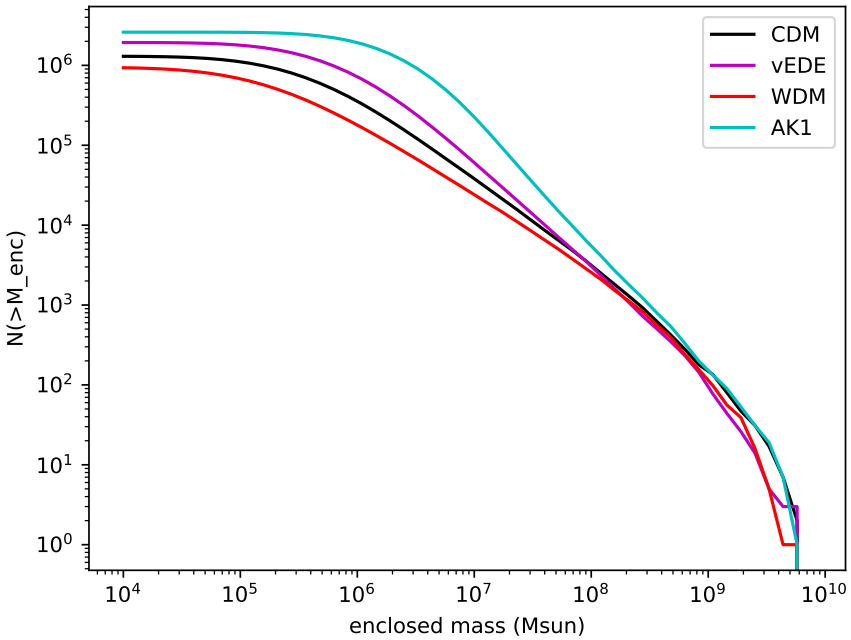

Figure 7: Complementary CDF of the enclosed mass $M_{\rm enc}(r_{1/2})$ for each model based on a sample of halos with $M > 10^8\ M_\odot/h$ and $r_{1/2} > 0.01\,$kpc. The halo sample for each power spectrum has the same $N(M)$ for $M \gtrsim 10^{10} M_\odot/h$, as seen in Fig. 5. Differences between curves indicate the observable impact of modified power spectra on subhalo populations.

[4] B. Diemer and M. Joyce, Astrophys. J. **871**, no.2, 168 (2019) doi:10.3847/1538-4357/aafad6 [arXiv:1809.07326 [astro-ph.CO]].

[5] E. O. Nadler *et al.* [DES], Astrophys. J. **893**, 48 (2020) doi:10.3847/1538-4357/ab846a [arXiv:1912.03303 [astro-ph.GA]].